

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled Multicenter Phase 3 Study to Evaluate Efficacy, Safety and Tolerability of Ianalumab on Top of Standard-of-care Therapy in Patients With Systemic Lupus Erythematosus (SIRIUS-SLE 2) **Short Title/Acronym:** SIRIUS-SLE 2 **Protocol Number:** CVAY736F12302 **NCT Number:** NCT05624749 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase 3 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of ianalumab compared to placebo, given as a monthly subcutaneous (s.c.) injection on top of standard-of-care (SoC) treatment, by assessing the proportion of participants achieving the Systemic Lupus Erythematosus Responder Index (SRI-4) at Week 60. **Secondary Objectives:** To evaluate the safety, tolerability, maintenance of reduced corticosteroid doses, and achievement of low disease activity states (such as BICLA response and SRI-6).

2.2 Background & Rationale Systemic Lupus Erythematosus (SLE) is an autoimmune disease where the immune system attacks healthy tissues, causing inflammation and tissue damage in affected organs. Ianalumab is a fully human monoclonal antibody that targets the B-cell activating factor receptor (BAFF-R). By depleting B-cells that play a central role in SLE pathogenesis, ianalumab aims to improve SLE disease activity and lupus symptoms. This study assesses whether ianalumab provides superior clinical benefit when added to standard-of-care therapies compared to standard-of-care alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 280 participants (Actual enrollment: 288) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 21-Apr-2023 **Estimated Primary Completion:** 12-Apr-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants aged ≥ 12 years or older (or limited to ≥ 18 years in countries where pediatric inclusion is not allowed) at the time of screening. 2. Diagnosis of systemic lupus erythematosus meeting the 2019 EULAR/ACR SLE classification criteria at least 6 months prior to screening. 3. Elevated serum titers at screening of anti-nuclear antibodies $\geq 1:80$ as determined by a central laboratory with a SLE-typical fluorescence pattern.

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women; Women of child-bearing potential (WOCBP) not using highly effective methods of contraception while on study treatment and for 6 months after stopping. 2. Prior treatment with anakinra. 3. History of receiving high dose corticosteroids, calcineurin inhibitors, JAK or other kinase inhibitors within 12 weeks; cyclophosphamide or biologics within 24 weeks; or belimumab within 12 weeks prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Anakinra 300 mg or matching placebo administered once monthly. **Route of Administration:** Subcutaneous (s.c.) injection via a pre-filled syringe. **Duration of Treatment:** 60 weeks of treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care (SoC) treatments for SLE, such as low-dose corticosteroids, anti-malarial medications, and standard immunosuppressants. **Prohibited:** High-dose corticosteroids, calcineurin inhibitors, JAK inhibitors, cyclophosphamide, belimumab, and other biological immunomodulators during the study timeframe.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Up to 4 Weeks Informed Consent, Medical History, EULAR/ACR Criteria Check, ANA Labs
2	Baseline	Day 0 Randomization, First s.c. Dose, Baseline SLEDAI-2K & BILAG Assessments
3	Interim	Weeks 4 to 56 Monthly s.c. Dosing, Safety Labs, Efficacy Assessments (SRI-4, Corticosteroid Tapering)
4	End of Study	Week 60 Primary Endpoint Assessment (SRI-4), IP Accountability, Transition to Follow-up/Extension

(Note: The 60-week treatment period is followed by a 20-week to 2-year safety follow-up period, or participants may enter an extension study.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants achieving Systemic Lupus Erythematosus Responder Index-4 (SRI-4) at Week 60. **Measurement Method:** Composite clinical response evaluating reductions in SLEDAI-2K score without worsening in BILAG-2004 or Physician's Global Assessment (PhGA).

7.2 Secondary & Exploratory Endpoints 1. Proportion of participants achieving a BICLA (BILAG-Based Composite Lupus Assessment) response. 2. Proportion of participants achieving an SRI-6 response. 3. Maintenance of reduced corticosteroid doses and achievement of low disease activity states.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical and laboratory monitoring, including vital signs and detection of anti-ianalumab antibodies. **Serious Adverse Events (SAEs):** Safety profile tracking, requiring standard 24-hour reporting mechanisms. **Data Monitoring Committee (DMC):** External and internal monitoring committees established by Novartis to continuously evaluate participant safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target $N=280$ powered sufficiently to detect statistically significant superiority of ianalumab + SoC vs. Placebo + SoC for the SRI-4 response at Week 60. **Handling of Missing Data:** Participants who withdraw early or require prohibited rescue medications are generally handled via Non-Responder Imputation (NRI) for binary composite endpoints (SRI-4, BICLA).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of investigator sites to ensure protocol adherence and patient safety. **Audit Trail:** Strict eCRF locking processes and source data verification by Novartis clinical operations teams.

11. Study Identifiers & Governance

Primary ID: CVAY736F12302 **Registry Number:** NCT05624749 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** SIRIUS-SLE 2 **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Multicenter Phase 3 Study to Evaluate Efficacy, Safety and Tolerability of Ianalumab on Top of Standard-of-care Therapy in Patients With Systemic Lupus Erythematosus

12. Clinical Framework (SLE Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE) **Diagnosis Threshold:** 2019 EULAR/ACR criteria & ANA \$\\ge 1:80\$ **Medical Methodology:** Standard-of-Care (SoC) Treatment + targeted biologic therapy **Optimization Target:** BAFF-R targeting / B-cell depletion

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for active SLE, integrating regular SoC with monoclonal antibody injections. **Education Protocol (HCP):** Training on composite SLE scoring indices (SLEDAI-2K, BILAG-2004, PhGA). **Education Protocol (Patient):** Education regarding s.c. injection procedures and AE self-reporting. **Quality Audit Mechanism:** Centralized laboratory tracking for ANA titers and anti-ianalumab antibodies; structured adherence monitoring.

14. Observation & Follow-Up Schedule

Total Duration: 80 to 164 weeks (60 weeks Treatment Period + 20 weeks to 2 years Follow-up Period) **Onsite Visit Cadence:** Monthly alignment for s.c. drug administration and ongoing clinical assessment. **Remote Monitoring Frequency:** Intermittent remote check-ins per standard site SOPs. **Checklist Review Interval:** Regular composite endpoint reviews (SRI-4/BICLA parameters).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Insert Name]	_____	06-Apr-2026				
Clinical Operations Lead	[Insert Name]	_____	06-Apr-2026				
Lead Statistician	[Insert Name]	_____	06-Apr-2026				